FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Meningitis

Tutorial | Week 3 · Fits, TB, kidneys & nutrition | Paediatrics | Dr Nabatte

Bacterial meningitis is a time-critical emergency in which the signs depend on age (in babies, fever, irritability and a bulging fontanelle rather than neck stiffness); do a lumbar puncture when it is safe, but never let it delay ceftriaxone, and read the CSF against age-specific norms while keeping TB and cryptococcal meningitis in mind for the subacute or HIV-positive child.

Must know before the ward round

1. Organisms by age: neonate (under 1 month) GBS, E. coli/Klebsiella, Listeria (rare in Uganda); after 1 month S. pneumoniae, N. meningitidis, Hib (now rare with pentavalent vaccine); plus non-typhoidal Salmonella in infants, HIV, malaria and sickle cell disease.
2. Neonates and infants rarely show neck stiffness: look for fever or hypothermia, poor feeding, irritability, high-pitched cry, apnoea, seizures and a bulging fontanelle (a late sign).
3. Give antibiotics within 1 hour of arrival (NICE NG240, 2024). Do the LP first only if it is safe and causes no meaningful delay; otherwise treat and tap later.
4. Do not do an LP with: signs of raised ICP (falling or GCS ≤9 / Blantyre ≤2, unequal or poorly reactive pupils, posturing, focal signs, Cushing triad), shock, respiratory failure, uncontrolled seizures, extensive purpura or bleeding risk (platelets low), or skin infection at the site.
5. Normal CSF after 1 month: WBC ≤5/µL (no neutrophils), protein <0.4 g/L, CSF:blood glucose ≥0.6. Neonate: WBC up to about 20/µL, protein up to 1.0 g/L.
6. Bacterial CSF: turbid, neutrophils 100–10,000/µL, protein >1 g/L, CSF:blood glucose <0.4 (WHO: glucose <1.5 mmol/L). TB: lymphocytes, protein 1–5 g/L, very low glucose. Crypto: few cells, CrAg/India ink positive, high opening pressure.
7. Empirical treatment over 1 month (UCG 2023): ceftriaxone 100 mg/kg/day IV/IM (once daily or 50 mg/kg 12-hourly; max 4 g/day) for 10–14 days. Alternative: chloramphenicol 25 mg/kg 6-hourly (plus ampicillin or benzylpenicillin).
8. Neonatal meningitis: ampicillin (50–100 mg/kg per dose, 12-hourly in week 1 then 8-hourly) plus gentamicin, or cefotaxime; treat for 21 days (UCG). Avoid ceftriaxone in jaundiced or preterm neonates and with calcium-containing fluids.
9. Dexamethasone 0.15 mg/kg IV 6-hourly for 4 days, first dose with or before the antibiotic (NICE: children over 3 months; continue only for pneumococcus or Hib). WHO Pocket Book: not routine in low-income settings; never in neonates or suspected cerebral malaria. UCG 2023 does not list it.
10. Contacts of meningococcal disease: ciprofloxacin 10 mg/kg single dose (max 500 mg) or ceftriaxone single IM dose (UCG 150 mg under 12 years), or rifampicin 10 mg/kg 12-hourly for 2 days.
11. Hearing loss is the commonest sequela (most often after pneumococcal meningitis): arrange audiology within 4 weeks of recovery; also watch for subdural effusion, hydrocephalus, SIADH and epilepsy.
12. TB meningitis: UCG 2RHZE/10RH (12 months) plus steroids; WHO 2022 also allows a 6-month intensive regimen (6HRZEto). Cryptococcal meningitis (HIV): induction with amphotericin B plus flucytosine plus fluconazole, therapeutic LPs, and delay ART 4–6 weeks.

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In a nutshell

Meningitis is inflammation of the meninges (the membranes around the brain and spinal cord). **Bacterial meningitis** kills or disables a large proportion of affected children within days, even with antibiotics, and in Uganda it is a **notifiable disease** (UCG 2023).

Three ideas carry the whole topic:

1. **The signs depend on age.** A 5-year-old has headache, neck stiffness and photophobia. A 3-week-old has only fever (or a low temperature), poor feeding, irritability and perhaps a seizure.
2. **Antibiotics first, investigations second.** The LP is the key test, but treatment must start **within 1 hour** of arrival, and an unsafe LP can kill.
3. **Read the CSF like a detective.** Cell type, protein and glucose separate bacterial, viral, TB and cryptococcal meningitis, which need completely different treatments.

> **CLINICAL PEARL**
> The one thing never to forget: **any febrile child with a convulsion, a bulging fontanelle, drowsiness or neck stiffness has meningitis until proven otherwise**. A positive malaria test does **not** exclude it.

Definitions & classification

Term	Meaning
Bacterial (pyogenic) meningitis	Acute bacterial infection of the meninges; CSF neutrophilic with low glucose
Neonatal meningitis	Meningitis in the first 28 days; **early-onset** (first 72 h, maternal flora: GBS, *E. coli*) vs **late-onset** (after 72 h, environmental/nosocomial: *Klebsiella*, other Gram-negatives, *S. aureus*)
Aseptic meningitis	Meningeal inflammation with negative bacterial cultures: viral (enteroviruses, HSV, HIV, mumps), partially treated bacterial, TB, fungal, drugs
Meningoencephalitis / encephalitis	Brain parenchyma involved: altered behaviour, focal signs, seizures (HSV, other viruses)
Tuberculous meningitis (TBM)	Subacute (1–3 weeks) *M. tuberculosis* meningitis; basal exudate, hydrocephalus, cranial nerve palsies
Cryptococcal meningitis	*Cryptococcus neoformans* meningitis in severe immunosuppression (advanced HIV, CD4 <100); high CSF opening pressure
Meningococcal disease	*N. meningitidis* meningitis and/or septicaemia (purpuric rash, shock)

Organisms by age

Age	Common organisms	Uganda notes
0–28 days	**Group B streptococcus**, ***E. coli***, ***Klebsiella*** and other Gram-negatives, *S. pneumoniae*, *Listeria* (rarely isolated in Africa), *S. aureus*	Late-onset Gram-negative infection (often resistant) is common in crowded neonatal units

Age	Common organisms	Uganda notes
1–3 months	Mixture of neonatal organisms and *S. pneumoniae*, *Hib*, *N. meningitidis*	Cover both groups
3 months–5 years	***S. pneumoniae***, ***N. meningitidis***, *Hib* (much less since vaccination), **non-typhoidal *Salmonella***	NTS meningitis: infants, malaria, HIV, SCD, malnutrition
Over 5 years	*S. pneumoniae*, *N. meningitidis*	Meningococcal epidemics in the dry season in the north
HIV-positive / malnourished	Pneumococcus, NTS, **TB**, ***Cryptococcus*** (older children and adolescents)	Ask for HIV status in every case

Epidemiology & Uganda context

- **Vaccines have changed the picture. Hib** vaccine (part of the pentavalent DTP-HepB-Hib at 6, 10 and 14 weeks) and **PCV** (pneumococcal conjugate, 6, 10 and 14 weeks) have greatly reduced Hib meningitis and lowered pneumococcal meningitis. **Check the card**: an unvaccinated infant is at much higher risk.
- **Meningococcus**: northern and north-western Uganda (West Nile, Acholi, Karamoja) lie at the edge of the **African meningitis belt**, where epidemics of *N. meningitidis* occur in the hot dry season.
- **Case fatality** for bacterial meningitis in African children is high (often 20–40% for pneumococcal disease), and about a third of survivors have sequelae (hearing loss, epilepsy, developmental delay).
- **TB and cryptococcal meningitis** follow the HIV epidemic: TBM peaks in children under 5 (often unvaccinated or with a household TB contact); cryptococcal meningitis is mainly seen in **older children and adolescents with advanced HIV**.

UGANDA CONTEXT

What is realistic. HC IIIs and HC IVs cannot usually do CSF analysis. They must recognise meningitis, **give a pre-referral dose of ceftriaxone (100 mg/kg IM)**, treat seizures and hypoglycaemia, and refer (UCG 2023). At hospitals, LP is done on the ward. CSF cell count, Gram stain, protein, glucose and India ink/CrAg are usually available at RRH level. Culture and GeneXpert Ultra on CSF are available at Mulago, the RRH labs and some mission hospitals (Mengo, Lacor). **CT is rarely available urgently** outside Kampala and a few RRHs, so the decision about LP is clinical.

Pathophysiology

1. **Colonisation** of the nasopharynx (pneumococcus, meningococcus, Hib) or, in neonates, exposure to maternal genital flora.
2. **Invasion into the blood** (bacteraemia). Polysaccharide **capsules** help organisms evade complement and phagocytosis, which is why **capsular conjugate vaccines** work and why **asplenia/SCD** (no splenic filtering) increases risk.
3. **Crossing the blood–brain barrier** at the choroid plexus; CSF has few antibodies and little complement, so bacteria multiply freely.
4. **Inflammation**: bacterial cell-wall products trigger cytokines (TNF, IL-1). Neutrophils pour in, the blood–brain barrier leaks (**protein rises**), bacteria and neutrophils consume glucose and CSF transport is impaired (**glucose falls**).
5. **Damage**: **vasogenic, cytotoxic and interstitial cerebral oedema → raised intracranial pressure**; **vasculitis and thrombosis** → infarcts (focal signs, seizures); exudate around the base and aqueduct → **hydrocephalus**; inflammation around the cochlea (labyrinthitis) → **sensorineural deafness**; inappropriate ADH secretion → **hyponatraemia**.

6. **Why dexamethasone might help:** antibiotic-induced lysis of bacteria releases more inflammatory material in the first hours; a steroid given **with or before** the first antibiotic damps this burst. That is why timing matters and why it helps most where the inflammation-driven damage (hearing loss with Hib and pneumococcus) dominates.

Clinical acumen

Focused history

Question	Why it matters
Duration of fever and illness (hours vs 1–3 weeks)?	Acute (hours–days) = bacterial or viral; **subacute (over 1 week)** = TB or cryptococcal
Convulsions: number, focal or generalised, duration?	Focal or prolonged seizures, or seizures outside the 6 months–5 years age range, are not "simple febrile convulsions"
Irritability, high-pitched cry, refusing feeds, vomiting, drowsiness?	The main signs in infants
Headache, neck pain, photophobia (older child)	Classic meningism
Rash (purpuric, spreading)	Meningococcal septicaemia: give ceftriaxone **now**
Antibiotics already taken	Partially treated meningitis: CSF culture often negative, cells still raised
Immunisation card: pentavalent (Hib), PCV, BCG scar	Unvaccinated = higher risk of Hib/pneumococcus; no BCG raises TBM concern
TB contact in the household, weight loss, chronic cough	TBM
HIV status (child and mother), on ART? CD4/viral load? Cotrimoxazole?	Crypto, TB, NTS; ART timing
Ear discharge, sinus pain, head injury, CSF leak, VP shunt	Pneumococcal source; recurrent meningitis
Neonate: maternal fever, PROM >18 h, GBS, preterm, LBW, birth asphyxia	Early-onset sepsis/meningitis risk
Sickle cell disease, malnutrition	Pneumococcus/NTS; altered fluid and antibiotic plans
Similar cases in the village/school (dry season, north)	Meningococcal epidemic: contact prophylaxis, notification

Examination

1. **ETAT**: airway, breathing, circulation (**shock** in meningococcal sepsis), coma (**AVPU/Blantyre/GCS**), convulsions, glucose.
2. **Vital signs** for age, temperature, SpO_2, **BP and heart rate together** (hypertension + bradycardia + irregular breathing = **Cushing triad**, late raised ICP).
3. **Skin**: **non-blanching petechiae or purpura** (glass test); check the whole body including soles.
4. **Fontanelle** (sitting quietly, not crying): **bulging** = raised ICP. **Head circumference** (baseline for hydrocephalus/subdural effusion).
5. **Meningism** (older than about 1 year): **neck stiffness** (chin cannot touch chest), **Kernig's sign** (pain/resistance on extending the knee with hip flexed), **Brudzinski's sign** (neck flexion causes hip and knee flexion). Absent in up to half of infants.
6. **Neurology**: pupils (size, symmetry, reaction), **cranial nerves** (VI palsy in TBM, VII), **focal limb weakness**, tone, posturing, fundi (papilloedema is uncommon early).

7. **Sources**: ears (otitis media, mastoiditis), sinuses, chest, skin, spine (dermal sinus in recurrent meningitis).
8. **Growth and nutrition**: weight, length, MUAC, oedema; **BCG scar**; lymph nodes, hepatosplenomegaly, oral thrush (HIV).

Red flags

RED FLAG

- **Purpuric or spreading petechial rash** with fever = meningococcal disease: **ceftriaxone immediately**, treat shock
- **Shock** (cold hands, CRT >3 s, weak fast pulse)
- **Signs of raised ICP**: falling conscious level (GCS ≤9 or Blantyre ≤2), unequal/sluggish pupils, decorticate/decerebrate posturing, focal signs, **bradycardia with hypertension**, irregular breathing, bulging fontanelle in a drowsy infant
- Prolonged or repeated seizures
- **Apnoea** in a neonate
- Hypoglycaemia; hyponatraemia (Na <130 mmol/L)

Thinking like a clinician

- **"Positive RDT + seizure + drowsy"** → both malaria and meningitis are possible. If the LP is safe, do it; if not, **treat both** (artesunate + ceftriaxone).
- **"Febrile convulsion" or meningitis?** A simple febrile convulsion is a **single, brief (<15 min), generalised** seizure in a child **6 months–5 years**, who is **fully alert within an hour**. Anything else (focal, prolonged, repeated, slow recovery, age under 6 months, partly treated with antibiotics, signs of meningism) → **LP**. Under 12 months, signs are unreliable, so the threshold for LP is low.
- **Subacute illness, cranial nerve palsy, weight loss, TB contact → TBM** until proven otherwise; don't wait for a positive culture.
- **HIV-positive adolescent with 2 weeks of headache and few signs → cryptococcal meningitis**: do serum/CSF CrAg and measure opening pressure.
- **"The CSF is clear, so it can't be meningitis."** Wrong: early bacterial, TB, crypto and viral CSF can all be clear. Send it.
- **Common student mistakes**: waiting for the LP before antibiotics; doing an LP on a shocked or deeply comatose child; forgetting glucose (blood glucose **just before LP** for the ratio); calling a neonate "well" because there is no neck stiffness; forgetting the hearing test and head circumference; giving ceftriaxone with calcium-containing fluid in a neonate.

EXAM FAVOURITE

Viva favourite: **"When would you NOT do a lumbar puncture?"** Answer by category: **raised ICP** (GCS ≤9 or falling, unequal/fixed pupils, posturing, focal signs, papilloedema, Cushing triad), **cardiorespiratory instability** (shock, respiratory failure), **ongoing seizures**, **bleeding risk** (extensive purpura, platelets low, DIC), and **local infection** at the LP site. In each case, **treat first** and do the LP when it becomes safe.

Clinical workup

Step 0: First 5 minutes

1. **ETAT triage** → emergency signs (airway, breathing, shock, coma, convulsions) to the resuscitation room.

2. **Airway and oxygen**; recovery position if unconscious; stop seizures (**diazepam 0.5 mg/kg PR or 0.2–0.3 mg/kg IV**, max 10 mg).
3. **IV/IO access**: blood for **glucose (bedside), blood culture, FBC, malaria RDT, electrolytes**, cross-match if pale or bleeding.
4. **Treat hypoglycaemia** (10% dextrose 5 mL/kg; 2 mL/kg in neonates).
5. **Shock** → 10–20 mL/kg crystalloid and reassess (not in SAM).
6. **Decide on LP now or later** (contraindications above). If it will delay treatment → **give ceftriaxone now** (and dexamethasone if your unit uses it) and do the LP when safe.
7. **Antibiotic within 1 hour** of arrival.

Step 1: Focused history

- [ ] **Onset and duration** → acute bacterial/viral vs subacute TB/crypto.
- [ ] **Seizures** (type, number, duration) → simple febrile convulsion vs meningitis/encephalitis.
- [ ] **Feeding, vomiting, irritability, cry, apnoea** (infant) → infant meningitis signs.
- [ ] **Rash** → meningococcal.
- [ ] **Prior antibiotics** → partially treated: rely on cells, protein, glucose and PCR.
- [ ] **Immunisation** (pentavalent, PCV, BCG) → organism likelihood; catch-up later.
- [ ] **HIV** (child, mother), **TB contact**, **SCD** → TB, crypto, NTS, pneumococcus.
- [ ] **Neonatal and maternal history** (PROM, maternal fever, GBS, prematurity) → early-onset organisms.
- [ ] **Development and head growth** before the illness → baseline for later sequelae.

Step 2: Focused examination

- **General**: temperature, weight, MUAC, Z-scores, hydration, pallor, jaundice, **rash** (glass test), BCG scar.
- **Neurological**: AVPU/Blantyre/GCS, **fontanelle**, **head circumference**, neck stiffness, Kernig, Brudzinski, pupils, cranial nerves (VI, VII, III), limb power and tone, reflexes, posturing, fundi.
- **Systemic**: ears, chest, heart, abdomen (spleen), lymph nodes, spine.
- **Document negatives**: "no petechiae, fontanelle soft, no focal neurology, pupils equal and reactive".

Step 3: Bedside tests

Test	How	What it tells you
Blood glucose	Glucometer before LP	Hypoglycaemia; needed for the CSF:blood glucose ratio
Malaria RDT	Finger prick	Co-infection; does not exclude meningitis
CSF appearance	Hold the tube against print	Turbid (bacterial), cobweb clot on standing (TB), blood-stained (traumatic tap or haemorrhage)
CSF glucose with a glucometer / dipstick	If the lab is closed: urine dipstick on CSF for leucocytes and glucose	Rough guide only; send to the lab
Serum CrAg lateral flow assay	Finger-prick or serum	Positive in cryptococcal disease (often before meningitis); do in HIV-positive children with headache
Opening pressure (manometer)	Child lying on the side, relaxed	Normal <20 cm H_2O; >25 cm H_2O in crypto → therapeutic LP
SpO_2, Hb (HemoCue)		Hypoxia, anaemia

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
CSF cell count and differential	Every LP	Neutrophils (bacterial), lymphocytes (viral, TB, crypto)	Diagnosis; repeat LP if not improving. RRH/NRH and many general hospitals
CSF protein and glucose (with paired blood glucose)	Every LP	Protein >1 g/L, ratio <0.4 in bacterial; very low glucose in TB	Diagnosis
CSF Gram stain	Every LP	Gram-positive diplococci (pneumococcus), Gram-negative diplococci (meningococcus), Gram-negative rods (Hib, *E. coli*, *Klebsiella*, NTS)	Guides antibiotic choice; negative in up to 60%
CSF culture and sensitivity	Every LP, ideally before antibiotics	Organism + resistance (e.g. ESBL Gram-negatives in neonates)	Change antibiotic; duration. RRH/NRH, Mengo, Lacor
CSF India ink, CrAg	HIV-positive, subacute, lymphocytic CSF	Encapsulated yeasts; CrAg positive (CrAg is more sensitive)	Cryptococcal treatment. Most RRHs
CSF GeneXpert MTB/RIF Ultra, ZN stain	Suspected TBM	MTB detected (sensitivity moderate; negative does not exclude)	Start TB treatment; rifampicin resistance. GeneXpert at most HC IVs and hospitals (via hub transport)
CSF PCR / latex agglutination	Prior antibiotics, culture negative	Pneumococcal, meningococcal, Hib antigen/DNA	Organism-specific duration; surveillance. National reference labs (UVRI, CPHL)
Blood culture	All, before antibiotics if possible	Positive in many bacterial cases	Organism when LP not possible
FBC, CRP	All	High WBC/neutrophils, low platelets (meningococcal DIC)	Supports bacterial cause; LP safety (platelets)
Electrolytes (Na), urea, creatinine	All admitted	**Hyponatraemia** (SIADH), AKI	Fluid plan
HIV test, CD4	All with unknown status	Positive; CD4 <100 → crypto	ART, crypto, TB plan
Chest X-ray	Suspected TB, pneumonia	Miliary pattern, hilar nodes	Supports TBM
Cranial ultrasound (open fontanelle)	Infants with enlarging head, persistent fever, focal signs	Ventriculomegaly, subdural effusion, abscess	Neurosurgical referral. Most hospitals
CT/MRI brain	Focal signs, suspected abscess/empyema, hydrocephalus, TBM complications, before LP if raised ICP is suspected (but treat first)	Hydrocephalus, infarcts, basal enhancement (TBM), abscess	Neurosurgery (VP shunt at Mulago, CURE Mbale). Kampala and some RRHs
Audiology (OAE/ABR)	All survivors of bacterial meningitis	Sensorineural hearing loss	Hearing aids, cochlear implant referral, school support. Mulago, Mengo ENT

Step 5: Putting it together

CSF interpretation table

	Appearance	WBC (/µL)	Main cell	Protein (g/L)	Glucose (CSF:blood)	Extra
Normal, over 1 month	Clear	**0–5**	Lymphocytes (no neutrophils)	**<0.4**	**≥0.6** (≥2.5 mmol/L)	Opening pressure <20 cm H_2O
Normal, neonate	Clear or slightly yellow	**up to about 20**	Lymphocytes (a few neutrophils possible)	**up to 1.0** (higher in preterms)	≥0.6	
Bacterial	**Turbid**	**100–10,000+** (may be low very early or in overwhelming sepsis)	**Neutrophils**	**>1.0**	**<0.4** (glucose often <1.5 mmol/L)	Gram stain, culture, antigen
Partially treated bacterial	Clear or turbid	Raised	Neutrophils or mixed	Raised	Low or normal	**Culture often negative**
Viral	Clear	**10–1,000**	**Lymphocytes** (neutrophils in the first 24–48 h)	0.4–1.0	**Normal**	Child usually less sick
TB	Clear, **cobweb clot** on standing	**50–500** (up to 1,000)	**Lymphocytes** (neutrophils early)	**1–5 (high)**	**<0.3, very low**	ZN rarely positive; Xpert Ultra
Cryptococcal	Clear	Often **<50** (may be near 0 in advanced HIV)	Lymphocytes	Normal to mildly raised	Normal to low	**India ink, CrAg positive; high opening pressure**
Cerebral malaria	Clear	Normal		Normal	Normal	Opening pressure may be raised

Model one-line summary

> "An 8-month-old boy, 7.6 kg (weight-for-age Z –1.2), with incomplete immunisation (no PCV3 or pentavalent 3), presenting with 2 days of fever, irritability, refusal to breastfeed and one focal seizure, with a bulging fontanelle and turbid CSF showing 2,400 WBC/µL (90% neutrophils), protein 2.1 g/L and CSF:blood glucose 0.15, Gram-positive diplococci: acute bacterial (likely pneumococcal) meningitis."

EXAM FAVOURITE

Worked example 1: the CSF glucose ratio and a traumatic tap. A 3-year-old's CSF glucose is **2.4 mmol/L** while the blood glucose taken just before the LP is **8.0 mmol/L** (he got dextrose in the ambulance). The absolute CSF glucose looks "almost normal", but the **ratio is 2.4 ÷ 8.0 = 0.3**, which is **low**: bacterial or TB until proven otherwise. That's why you measure blood glucose immediately before the LP. A second sample is blood-stained: **RBC 50,000/µL, WBC 120/µL**. A rough rule allows about **1 WBC per 500–1,000 RBC** (so about 50–100 WBC are explained by blood), but correction formulas are unreliable: if the child looks unwell, **treat as meningitis** and send for culture.

EXAM FAVOURITE

Worked example 2: telling TB from bacterial. A 4-year-old with 12 days of fever, vomiting and a right **VI nerve palsy**: CSF clear, WBC **280/µL (80% lymphocytes)**, protein **2.8 g/L**, glucose ratio **0.2**, Gram stain negative. Lymphocytes + very high protein + very low glucose + subacute course + cranial nerve palsy = **TB meningitis**. Start TB treatment and steroids today; do not wait for Xpert or culture.

Differential diagnosis

Condition	Distinguishing features	Key investigation
Cerebral malaria	Coma with parasitaemia, normal CSF, retinopathy; may coexist	RDT/slide, LP
Simple febrile convulsion	6 months–5 years, single brief generalised seizure, fully alert within 1 h, no meningism	Clinical; LP if any doubt (especially under 12 months)
Viral encephalitis (HSV)	Behaviour change, focal seizures, reduced consciousness, lymphocytic CSF with RBCs	CSF PCR (rarely available), MRI/EEG; start aciclovir if suspected and available
TB meningitis	Subacute, TB contact, weight loss, cranial nerve palsies, hydrocephalus	CSF (lymphocytes, high protein, very low glucose), Xpert, CXR
Cryptococcal meningitis	HIV with low CD4, headache over 1–2 weeks, few signs, high opening pressure	Serum and CSF CrAg, India ink
Brain abscess / subdural empyema	Focal signs, persistent fever, sinusitis/otitis, cyanotic heart disease	CT/MRI (LP contraindicated)
Sepsis without meningitis	Shock, no meningism; LP normal	Blood culture, LP when stable
Pneumonia with meningismus	Upper-lobe pneumonia can cause neck stiffness	Chest exam/CXR; CSF normal
Hypoglycaemia, poisoning, Reye-type encephalopathy	History, no fever or meningism	Glucose, toxicology history
Intracranial haemorrhage (trauma, vitamin K deficiency in infants)	Sudden onset, pallor, bulging fontanelle, history of no vitamin K	Cranial US/CT, clotting
Retropharyngeal abscess, cervical lymphadenitis	Neck pain/stiffness, drooling	ENT exam, lateral neck X-ray

Investigations

The full table with Ugandan availability is in **Clinical workup, Step 4**. Core principles:

1. **Blood culture and glucose before antibiotics**, but never delay antibiotics for them.
2. **LP before antibiotics only if it is safe and quick.** CSF cells, protein and glucose are not much affected by antibiotics given in the previous few hours (within about 24 h); culture is.
3. **Always request the CSF basics together**: cell count and differential, protein, glucose (with paired blood glucose), Gram stain, culture; add **CrAg/India ink** and **Xpert Ultra/ZN** when the history suggests HIV or TB.
4. **Repeat the LP** after 48–72 h if the child is not improving, the organism is unusual (Gram-negative in a neonate) or resistant, or fever persists.

Management

A. Emergency algorithm (any age)

1. **ABC, oxygen, stop seizures, check glucose.**
2. **Shock?** Crystalloid 10–20 mL/kg, reassess, repeat once; inotropes only in HDU/ICU.
3. **Is LP safe now?** If yes and it adds no delay, do it; if not, go straight to step 4.
4. **Give the first dose of antibiotic within 1 hour** (dexamethasone first or together if indicated).
5. **Supportive care**: head up 30°, head in midline, fever control, maintenance fluids, NGT feeds if drowsy, monitor GCS/Blantyre, pupils, BP, HR, RR, Na, glucose.
6. **Review at 24–48 h** with CSF/blood culture results; narrow antibiotics; decide duration.
7. **Notify** (district surveillance focal person) and give **contact prophylaxis** for meningococcal disease and Hib.
8. **Before discharge**: neurological and **hearing assessment**, head circumference, development plan, immunisation catch-up, HIV/TB follow-up.

B. Empirical antibiotics by age

Infants over 1 month and children (UCG 2023 / WHO Pocket Book)

Drug	Dose	Route	Frequency	Duration / max	Notes
Ceftriaxone (first line)	**100 mg/kg/day**	IV (slow over 3–5 min, or infusion) or deep IM	**Once daily**, or **50 mg/kg 12-hourly**	**10–14 days** empirically (UCG); **max 4 g/day**	Same dose as pre-referral; good CSF penetration
Cefotaxime (alternative)	**50 mg/kg**	IV	**6-hourly**	7–14 days; max 12 g/day	Preferred in neonates
Chloramphenicol (if no ceftriaxone, or not improving and sensitive)	**25 mg/kg**	IV/IM (oral once improving)	**6-hourly**	10–14 days; max 1 g/dose	**Not under 1 month** (grey baby syndrome); bone-marrow suppression
+ Ampicillin (WHO PB, with chloramphenicol)	**50 mg/kg**	IV/IM	6-hourly	10 days	
+ or Benzylpenicillin (WHO PB, with chloramphenicol)	**100,000 IU/kg (60 mg/kg)**	IV/IM	6-hourly	10 days	

Organism-specific (when culture or Gram stain is known)

Organism	Treatment (UCG 2023 / NICE NG240)	Duration
S. pneumoniae	Ceftriaxone 100 mg/kg/day; or benzylpenicillin 100,000 IU/kg 4-hourly if sensitive	**10–14 days** (UCG; up to 21 if severe); NICE 10 days
N. meningitidis	Ceftriaxone, or benzylpenicillin 100,000–150,000 IU/kg 6-hourly, or chloramphenicol	**5–7 days** (NICE 5 days; UCG up to 14)
Hib	Ceftriaxone; switch to chloramphenicol or ampicillin only if the isolate is sensitive	**10 days**
GBS	Ceftriaxone/cefotaxime (or benzylpenicillin + gentamicin)	**14 days** (NICE); UCG neonatal 21 days
Gram-negative bacilli (*E. coli*, *Klebsiella*, NTS)	Cefotaxime/ceftriaxone if sensitive; **meropenem** (40 mg/kg 8-hourly) for ESBL	**At least 21 days**
Listeria	Ampicillin/amoxicillin IV	**21 days**

DRUG DOSING

Ceftriaxone safety: do **not** give it at the same time as (or in the same line as) **calcium-containing fluids** (Ringer's lactate, Hartmann's, calcium gluconate) in neonates (fatal precipitates; MHRA/NICE 2024), and avoid it in **jaundiced or preterm neonates** (it displaces bilirubin). Use **cefotaxime** in neonates where possible.

C. Neonatal meningitis (UCG 2023 / WHO Pocket Book)

1. **Supportive care**: keep warm (skin-to-skin/KMC if stable, warmer if not), prevent hypoglycaemia (breastfeed/expressed milk by NGT, or IV 10% dextrose maintenance 60 mL/kg/day on day 1 rising by 20 mL/kg/day), oxygen if SpO_2 <92% (UCG), treat seizures with **phenobarbital 20 mg/kg** IV loading (WHO PB; UCG 10–15 mg/kg), avoid paracetamol for fever (UCG: undress, control the environment).
2. **Antibiotics for 21 days** (UCG; GBS may be 14 days per NICE):

Drug	Dose	Route	Frequency	Notes
Ampicillin	**50–100 mg/kg** per dose (use the higher dose for meningitis)	IV	**12-hourly in week 1; 8-hourly from day 7**	Covers GBS, *Listeria*, *Enterococcus*
plus Gentamicin	**5 mg/kg** (term) or **3 mg/kg** (LBW/preterm) in week 1; **7.5 mg/kg** after week 1 (WHO PB)	IV/IM	**24-hourly**	UCG 2023 lists 2.5 mg/kg 12-hourly; check levels/renal function if prolonged. Consistent with the neonatal-resuscitation topic
or Cefotaxime (preferred where resistance or Gram-negative meningitis)	**50 mg/kg**	IV	12-hourly (week 1), 8-hourly (weeks 2–4)	Add ampicillin to cover *Listeria*/GBS empirically
Benzylpenicillin (GBS confirmed)	50,000–100,000 IU/kg (<7 days) every 8 h; 100,000–150,000 IU/kg every 4–6 h (older)	IV		Plus gentamicin

3. **Monitor**: temperature, feeding, weight, glucose, **head circumference daily**, fontanelle, seizures; repeat LP if Gram-negative or not improving.
4. **Before discharge**: hearing screen, cranial ultrasound if any concern, neurodevelopmental follow-up, BCG/OPV/HepB if missed.

D. Dexamethasone: the controversy

	Evidence and guidance
Rationale	Dampens the inflammatory burst when antibiotics lyse bacteria; reduces **hearing loss** (especially Hib)
Dose	**0.15 mg/kg IV 6-hourly for 4 days** (2–4 days; max 10 mg per dose); first dose **with or 10–20 min before** the first antibiotic; not if more than about 12 h after antibiotics started (NICE; RCH Melbourne prefers within 4 h)
NICE NG240 (2024)	Give IV dexamethasone to children **over 3 months** with strongly suspected or confirmed bacterial meningitis; **infection specialist advice for 28 days to 3 months**; continue only if the organism is **pneumococcus or Hib**; stop for others
Cochrane (2015)	Steroids reduced hearing loss and neurological sequelae overall, but the benefit was seen in **high-income countries**; **no benefit in low-income countries**

	Evidence and guidance
African trial data	A large randomised trial in **Malawian children** found **no benefit** of dexamethasone on mortality or sequelae; late presentation, HIV, malnutrition and pre-treatment probably explain this
WHO Pocket Book (2013)	Insufficient evidence for routine use in developing countries **except in TB meningitis**. **Do not use** in **neonates**, presumed **cerebral malaria**, presumed **viral encephalitis**, or where antibiotic resistance is suspected and steroids may mask failure
UCG 2023	**Not mentioned** in the meningitis section

UGANDA CONTEXT

What Ugandan wards actually do: most units **do not give dexamethasone routinely** to young children with bacterial meningitis, in line with the WHO Pocket Book and UCG, because most children present late, many have HIV, malnutrition or malaria, and African trials showed no benefit. Some consultants give it to **older children with suspected pneumococcal or Hib meningitis who present early and are HIV-negative**, mirroring NICE. **Always** give steroids in **TB meningitis**. Ask your consultant which policy the unit follows and say so in your presentation.

E. Supportive care

Problem	Management
Seizures	Diazepam 0.5 mg/kg PR or 0.2–0.3 mg/kg IV (max 10 mg), repeat once; **phenobarbital 15–20 mg/kg** IV loading (UCG 10–15 mg/kg), or phenytoin 15–20 mg/kg IV over 20 min; maintenance phenobarbital 5 mg/kg/day
Fever / pain	Paracetamol 15 mg/kg 6-hourly (not in neonates per UCG)
Fluids	**Full maintenance** (100/50/20 mL/kg/day), isotonic (0.9% saline with 5% dextrose or equivalent), ideally via NGT feeds when safe. **Restrict to about two-thirds** only if there is **SIADH** (Na <130 mmol/L, no dehydration) or signs of raised ICP; never restrict in shock or dehydration
Raised ICP	Head up 30°, neutral neck, treat seizures/fever/hypoglycaemia, avoid hypotonic fluids, ensure oxygenation; consider **mannitol 0.25–0.5 g/kg IV** over 20–30 min (or 3% saline 3–5 mL/kg) as a temporary measure (NICE: osmotic agents, not glycerol); paediatric critical care advice
Hyponatraemia	Usually SIADH: fluid adjustment; if Na <120 or seizures, 3% saline 2–3 mL/kg over 10–20 min
Nutrition	NGT feeds if drowsy; breastfeeding continues
Nursing	Turn 2-hourly, eye care, bladder care, monitor GCS/Blantyre, pupils, vitals 1–4-hourly, head circumference daily in infants

F. Meningococcal disease: extra points

- **Septicaemia ± meningitis**: purpuric rash, shock, DIC. **Ceftriaxone immediately** (pre-hospital IM if transfer is delayed), aggressive shock management (repeated boluses, inotropes in ICU), correct glucose, calcium and potassium, transfuse platelets/FFP for active bleeding.
- **Epidemic setting** (WHO Pocket Book): single-dose **ceftriaxone 100 mg/kg IM** (repeat if not improving at 24 h), 5 days for children 2 months–5 years and at least 7 days under 2 months.
- **Notify immediately**; **contact prophylaxis** (table below); index case also needs carriage eradication if treated with penicillin or chloramphenicol only.

Prophylaxis (close/household contacts, especially under 5)	Dose
Ciprofloxacin (UCG first choice)	Child **<12 years: 10 mg/kg** single oral dose (max 500 mg); ≥12 years: 500 mg
Ceftriaxone (e.g. pregnancy)	UCG: child <12 years **150 mg IM** single dose (BNFC 125 mg); ≥12 years 250 mg
Rifampicin (meningococcal)	**10 mg/kg** (5 mg/kg under 1 month) **12-hourly for 2 days**, max 600 mg/dose
Rifampicin (Hib household with an unvaccinated child under 4)	**20 mg/kg once daily for 4 days**, max 600 mg

G. TB meningitis (UCG 2023; WHO 2022)

1. **Start treatment on clinical and CSF grounds**; mortality and disability rise with every day of delay.
2. **Regimen**: UCG 2023: **2RHZE / 10RH** (12 months in total). The **WHO 2022 consolidated TB guidelines for children and adolescents** also recommend a **6-month intensive regimen (6HRZEto: isoniazid, rifampicin, pyrazinamide and ethionamide, at higher doses)** as an alternative for bacteriologically confirmed or clinically diagnosed TBM without suspected drug resistance. Follow your unit's TB programme for which regimen is stocked.
3. **Standard child doses** (WHO): **isoniazid 10 mg/kg** (7–15; max 300 mg), **rifampicin 15 mg/kg** (10–20; max 600 mg), **pyrazinamide 35 mg/kg** (30–40), **ethambutol 20 mg/kg** (15–25), given as child-friendly fixed-dose combinations by weight band.
4. **Steroids for all**: **dexamethasone 0.6 mg/kg/day** (WHO Pocket Book) or **prednisolone 2 mg/kg/day** (max 60 mg) for **4 weeks**, then taper over 2–4 weeks.
5. **Hydrocephalus**: cranial US/CT; neurosurgical referral for a VP shunt (Mulago, CURE Children's Hospital Mbale).
6. **HIV**: start ART **4–8 weeks** after TB treatment begins (TBM: delay to avoid IRIS), cotrimoxazole, pyridoxine with isoniazid.
7. **Contact tracing** and TB preventive therapy for household children.

MRC staging of TBM: **Stage I** alert, no focal signs; **Stage II** GCS 11–14 or focal signs (e.g. cranial nerve palsy); **Stage III** GCS ≤10. Outcome worsens with stage.

H. Cryptococcal meningitis (HIV)

- Mostly in **older children and adolescents** with advanced HIV (CD4 <100 or <15% in younger children). Screen with **serum CrAg**.
- **Induction (WHO 2022 recommendation for adults, adolescents and children)**: **single high-dose liposomal amphotericin B 10 mg/kg IV** + **flucytosine 100 mg/kg/day** (in 4 doses) + **fluconazole 12 mg/kg/day** (max 800 mg) for **14 days**. Alternative: amphotericin B deoxycholate 1 mg/kg/day + flucytosine for 7 days, then fluconazole.
- **Consolidation**: fluconazole **6–12 mg/kg/day** (max 800 mg) for 8 weeks; **maintenance** 6 mg/kg/day (max 200 mg) until immune recovery.
- **Raised pressure**: **therapeutic LPs** (remove CSF until the opening pressure is halved or below 20 cm H_2O), repeat daily while symptomatic. No steroids, no acetazolamide, no mannitol for crypto.
- **ART timing**: **delay ART 4–6 weeks** after starting antifungal treatment (UCG flowchart; COAT trial showed higher mortality with early ART).
- UCG 2023 says **refer to hospital**. Flucytosine and liposomal amphotericin are mainly at national and regional referral hospitals.

UGANDA CONTEXT

At HC IV level: if serum CrAg is positive and the child has headache or other signs, **refer for LP**. If CrAg is positive with no signs and LP is not possible, the UCG flowchart gives **pre-emptive fluconazole** (adult doses in UCG; children 12 mg/kg/day, check with the ART clinic). Do not start ART immediately in a child with suspected cryptococcal meningitis.

I. Setting-specific plan

Setting	Action
HC II/III	Recognise (IMCI danger signs, stiff neck, bulging fontanelle), **ceftriaxone 100 mg/kg IM (or IV) pre-referral**, diazepam for seizures, glucose (or sugar water), keep warm, refer with a note
HC IV	As above plus IV access, maintenance fluids, oxygen; LP if trained and safe (send CSF with the patient or to the hub lab); continue ceftriaxone; refer complicated cases
General hospital / RRH	LP, CSF analysis, blood culture, full course of antibiotics, seizure care, Xpert, CrAg, cranial US; notification
NRH (Mulago) / CURE Mbale	ICU, CT/MRI, neurosurgery (VP shunt, abscess drainage), audiology, rehabilitation

Complications

Timing	Complications
Acute	Septic shock, DIC (meningococcal), **raised ICP and herniation**, seizures/status epilepticus, **SIADH/hyponatraemia**, hypoglycaemia, aspiration, cerebral infarction (vasculitis, venous sinus thrombosis)
Subacute	**Subdural effusion** (common in infants, especially Hib/pneumococcal: persistent fever, enlarging head, bulging fontanelle; usually resolves), **subdural empyema**, **brain abscess**, **hydrocephalus** (communicating or obstructive), **ventriculitis** (neonates), cranial nerve palsies (III, IV, VI, VII), persistent or recurrent fever (drug fever, phlebitis, nosocomial infection, abscess)
Long-term	**Sensorineural hearing loss** (most frequent; up to a third after pneumococcal meningitis; bilateral profound in a few percent), **epilepsy**, **cerebral palsy/motor deficits**, cortical blindness, **developmental delay and learning difficulties**, behavioural problems; after meningococcal septicaemia: skin scars, amputations, bone growth problems

CLINICAL PEARL

Hearing loss is silent. NICE NG240 says offer **audiology within 4 weeks of the child being well enough** (preferably before discharge). Severe bilateral loss after pneumococcal meningitis can be followed by **cochlear ossification**, which makes implants harder later, so early testing matters.

Ward-round & exam pearls

Model case presentation

> "Baby K is an 8-month-old boy from Wakiso, **weight 7.6 kg (weight-for-age Z −1.2), length 68 cm (length-for-age Z −0.8), MUAC 13.4 cm**, admitted yesterday with 2 days of fever, irritability, refusal to breastfeed and a single focal seizure of the left arm lasting 10 minutes. He received **BCG, OPV0–3, pentavalent 1–2 and PCV 1–2** but missed the 14-week visit; measles-rubella is due at 9 months. He is **exclusively breastfed with porridge started at 6 months**. Mother is **HIV negative** (tested in this pregnancy). No TB contact. On admission: temperature 39.1 °C, RR 44, HR 168, CRT 2 s, SpO_2 96%, **bulging fontanelle**,

> irritable, no rash, pupils equal and reactive, no focal deficit after the seizure. RDT negative, glucose 4.6 mmol/L. **CSF turbid: WBC 2,400/µL (90% neutrophils), protein 2.1 g/L, glucose 0.8 mmol/L (blood 5.2; ratio 0.15), Gram-positive diplococci. Assessment: acute bacterial meningitis, probably pneumococcal, in a partially vaccinated infant.** On **ceftriaxone 100 mg/kg (760 mg) IV once daily**, day 2; no further seizures; feeding by breast and cup. Sodium 132 mmol/L. **Plan:** ceftriaxone for 10–14 days; maintenance fluids with daily sodium; head circumference daily (today 44 cm); culture and sensitivity pending; notify; **hearing test before discharge**; catch up pentavalent 3 and PCV 3; developmental follow-up."

Likely viva questions

1. *Signs of meningitis in an infant?* Fever or hypothermia, irritability, high-pitched cry, poor feeding, vomiting, lethargy, seizures, apnoea, bulging fontanelle (late); neck stiffness often absent.
2. *Contraindications to LP?* Raised ICP signs, shock/respiratory failure, uncontrolled seizures, bleeding risk/extensive purpura, skin infection at the site.
3. *Empirical antibiotic for a 2-year-old?* Ceftriaxone 100 mg/kg/day IV (once daily or 12-hourly), max 4 g/day.
4. *And for a 10-day-old?* Ampicillin + gentamicin (or ampicillin + cefotaxime), 21 days.
5. *CSF in TB meningitis?* Lymphocytes 50–500, protein 1–5 g/L, very low glucose, cobweb clot.
6. *When do you give dexamethasone?* NICE: over 3 months, with/before the first antibiotic, 0.15 mg/kg 6-hourly × 4 days; WHO/UCG do not recommend routine use in low-income settings; always in TBM.
7. *Most common long-term complication?* Sensorineural hearing loss.

OSCE tips: LP station: consent, position (lateral, knees to chest, held by an assistant), landmark (**L3/4 or L4/5**, intercristal line; in infants the cord ends lower, so use L4/5), aseptic technique, needle with stylet, measure opening pressure, collect 3 tubes (micro, biochemistry, extra), note the time and blood glucose. Neuro-exam station in an infant: always say "I would measure the head circumference and feel the fontanelle with the baby upright and calm."

Mnemonics & memory aids

- **LP contraindications: "SPIRAL"**: **S**hock, **P**latelets low/purpura, **I**CP raised (falling GCS, pupils, posturing, focal signs), **R**espiratory failure, **A**ctive seizures, **L**ocal skin infection.
- **CSF: "Bacteria eat sugar, TB eats more"**: bacterial = neutrophils + low glucose; TB = lymphocytes + very low glucose + very high protein; viral = lymphocytes + normal glucose.
- **Neonatal organisms: "GEL"**: **G**BS, ***E**. coli*, ***L**isteria*.
- **Older child: "PNH"**: **P**neumococcus, ***N**. meningitidis*, ***H**aemophilus*.
- **Dexamethasone "0.15 × 4 × 4"**: 0.15 mg/kg, 4 times a day, for 4 days, only with the first antibiotic.
- **Ceftriaxone "100, 4, no calcium"**: 100 mg/kg/day, max 4 g, never with calcium in neonates.

High-yield summary

Numbers & doses to know

Item	Value
Antibiotic timing	**Within 1 hour** of arrival (NICE NG240)
Ceftriaxone	**100 mg/kg/day** IV/IM once daily or 50 mg/kg 12-hourly; **max 4 g/day**; 10–14 days (UCG)
Cefotaxime	**50 mg/kg 6-hourly** (children); 12-hourly → 8-hourly in neonates
Chloramphenicol	**25 mg/kg 6-hourly** (max 1 g/dose); not under 1 month

Item	Value
Neonatal ampicillin	**50–100 mg/kg** 12-hourly (week 1), 8-hourly after
Neonatal gentamicin	**5 mg/kg** daily (term), **3 mg/kg** (LBW/preterm) in week 1; UCG 2.5 mg/kg 12-hourly
Neonatal duration	**21 days** (UCG); GBS 14 days (NICE)
Duration by organism	Meningococcus 5–7 d; Hib 10 d; pneumococcus 10–14 d; GBS 14 d; Gram-negative and *Listeria* 21 d
Dexamethasone	**0.15 mg/kg 6-hourly × 4 days**, first dose with/before antibiotic; NICE: over 3 months
TBM steroids	Dexamethasone 0.6 mg/kg/day or prednisolone 2 mg/kg/day (max 60 mg) × 4 weeks, then taper
Normal CSF (over 1 month)	WBC **≤5**, protein **<0.4 g/L**, glucose ratio **≥0.6**
Normal CSF (neonate)	WBC **up to ~20**, protein **up to 1.0 g/L**
Bacterial CSF	Neutrophils **100–10,000**, protein **>1 g/L**, ratio **<0.4** (glucose <1.5 mmol/L)
Opening pressure	Normal **<20 cm H_2O**; crypto often >25 → therapeutic LP
Diazepam	0.5 mg/kg PR / 0.2–0.3 mg/kg IV, max 10 mg
Phenobarbital	15–20 mg/kg IV loading
Mannitol	0.25–0.5 g/kg IV over 20–30 min
Meningococcal prophylaxis	Ciprofloxacin **10 mg/kg** once (max 500 mg); rifampicin **10 mg/kg** 12-hourly × 2 days
Hib prophylaxis	Rifampicin **20 mg/kg** daily × 4 days (max 600 mg)
Crypto induction	L-AmB **10 mg/kg** once + flucytosine **100 mg/kg/day** + fluconazole **12 mg/kg/day** × 14 days
ART after crypto	Delay **4–6 weeks**
Audiology	Within **4 weeks** of recovery

Classic exam traps

- **"No neck stiffness, so not meningitis"** in a 4-month-old. Wrong: infants often lack meningism; look at the fontanelle and behaviour.
- **"Do the LP first, then start antibiotics"** in a shocked or deeply comatose child. Wrong: **resuscitate and treat**, tap later.
- **"Positive RDT explains the seizure."** Wrong: do an LP (if safe) or cover meningitis.
- **"CSF is clear, so meningitis is excluded."** Wrong: send cells, protein, glucose, CrAg and Xpert.
- **"Give dexamethasone to every child with meningitis."** Wrong: never in neonates or cerebral malaria; not routine in low-income settings; always in **TBM**.
- **"Ceftriaxone with Ringer's lactate in a neonate."** Wrong: calcium precipitates; use cefotaxime.
- **"Start ART immediately in cryptococcal meningitis."** Wrong: wait **4–6 weeks**.
- **"Chloramphenicol is fine for neonates."** Wrong: grey baby syndrome.

Questions seniors ask

Why do you check the blood glucose just before the LP? To calculate the CSF:blood glucose ratio; a CSF glucose that looks normal can be low relative to a high blood glucose (after dextrose).

This child had amoxicillin for 3 days before admission. How will that change the CSF? Cells, protein and glucose stay abnormal, but Gram stain and culture are often negative; rely on the cell pattern and antigen/PCR tests and give a full course.

Would you give dexamethasone to this 8-month-old in Kampala? Explain both sides: NICE would give it with the first antibiotic to a child over 3 months; WHO Pocket Book and UCG do not recommend routine use because African trials showed no benefit; follow the unit policy. If it is given, it must be with or before the first antibiotic dose.

The fever returns on day 7 and the head is getting bigger. What do you think? Subdural effusion or empyema, hydrocephalus, or abscess (also phlebitis, drug fever, nosocomial infection). Measure the head circumference, do a cranial ultrasound (open fontanelle) or CT, and repeat the LP if needed.

How do you differentiate TB from bacterial meningitis at the bedside? TBM: subacute course (over a week), TB contact, weight loss, cranial nerve palsy, hydrocephalus, lymphocytic CSF with very high protein and very low glucose. Bacterial: acute, neutrophilic, turbid.

Who needs prophylaxis after a meningococcal case? Household and close contacts (especially under 5s) and anyone exposed to the patient's oral secretions: ciprofloxacin single dose (or ceftriaxone, rifampicin); the index case needs eradication if not treated with ceftriaxone.

What follow-up will you arrange? Hearing test within 4 weeks, developmental assessment, head circumference, seizure follow-up, immunisation catch-up (Hib, PCV), and HIV/TB care where relevant.

SELF-TEST

Flashcards

28 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Commonest organisms causing neonatal meningitis?	**Group B streptococcus, *E. coli* and other Gram-negatives (*Klebsiella*)**, *Listeria* (rarely isolated in Africa), *S. pneumoniae*, *S. aureus*. Mnemonic **GEL**.
2	Commonest organisms causing bacterial meningitis after 3 months?	***S. pneumoniae*, *N. meningitidis*, *Hib*** (rare since vaccination); in Uganda also **non-typhoidal *Salmonella*** (infants, HIV, malaria, SCD, malnutrition).
3	Signs of meningitis in a young infant?	Fever or hypothermia, **irritability**, high-pitched cry, **poor feeding**, vomiting, lethargy, **seizures**, apnoea, **bulging fontanelle** (late). Neck stiffness is often absent.
4	Kernig's and Brudzinski's signs?	**Kernig**: with hip flexed to 90°, extending the knee causes pain/resistance. **Brudzinski**: passive neck flexion causes involuntary hip and knee flexion. Both are unreliable under about 1 year.
5	Contraindications to lumbar puncture (SPIRAL)?	**S**hock, **P**latelets low/extensive purpura or bleeding risk, **I**CP raised (GCS ≤9 or falling, unequal/sluggish pupils, posturing, focal signs, Cushing triad), **R**espiratory failure, **A**ctive (uncontrolled) seizures, **L**ocal skin infection at the site.
6	Normal CSF in a child over 1 month?	Clear; **WBC 0–5/µL, no neutrophils**; **protein <0.4 g/L**; **CSF:blood glucose ≥0.6** (≥2.5 mmol/L); opening pressure <20 cm H2O.
7	Normal CSF in a neonate?	**WBC up to about 20/µL**, **protein up to 1.0 g/L** (higher in preterms), glucose ratio ≥0.6. Thresholds are higher than in older children.
8	Typical CSF in bacterial meningitis?	**Turbid**, **neutrophils 100–10,000/µL**, **protein >1 g/L**, **CSF:blood glucose <0.4** (WHO: glucose <1.5 mmol/L); Gram stain positive in about half.
9	Typical CSF in TB meningitis?	Clear with a **cobweb clot**, **lymphocytes 50–500/µL**, **protein 1–5 g/L (very high)**, **glucose very low (ratio <0.3)**; ZN rarely positive; send GeneXpert Ultra.
10	Typical CSF in cryptococcal meningitis?	Clear, **few cells (often <50, may be near 0)**, lymphocytes, normal to mildly raised protein, normal/low glucose, **India ink and CrAg positive**, **opening pressure high (>25 cm H2O)**.
11	Empirical antibiotic for meningitis in a child over 1 month (UCG 2023)?	**Ceftriaxone 100 mg/kg/day** IV/IM, once daily or 50 mg/kg 12-hourly, **max 4 g/day**, for **10–14 days**. Alternative: chloramphenicol 25 mg/kg 6-hourly.
12	Empirical antibiotics for neonatal meningitis (UCG 2023)?	**Ampicillin 50–100 mg/kg** (12-hourly week 1, 8-hourly after) **+ gentamicin** (UCG 2.5 mg/kg 12-hourly; WHO 5 mg/kg daily term, 3 mg/kg LBW), or cefotaxime + ampicillin; **21 days**.
13	Why avoid ceftriaxone in some neonates?	It **displaces bilirubin** (kernicterus risk in jaundiced/preterm babies) and forms **fatal precipitates with calcium-containing IV fluids** (e.g. Ringer's lactate). Use cefotaxime.
14	Dexamethasone dose and timing in bacterial meningitis?	**0.15 mg/kg IV 6-hourly for 4 days** (2–4 days), first dose **with or 10–20 min before** the first antibiotic; not if >12 h after antibiotics. NICE: over 3 months; continue only for pneumococcus or Hib.
15	Where is dexamethasone NOT recommended (WHO Pocket Book)?	**Neonates**, presumed **cerebral malaria**, presumed **viral encephalitis**, and routine use in low-income settings (no benefit shown). **Always give steroids in TB meningitis.**

#	Question	Answer
16	Meningococcal contact prophylaxis doses (child)?	**Ciprofloxacin 10 mg/kg** single oral dose (max 500 mg) (UCG first choice); or **ceftriaxone** single IM dose (UCG 150 mg under 12 years); or **rifampicin 10 mg/kg** (5 mg/kg under 1 month) 12-hourly × 2 days (max 600 mg).
17	Hib household prophylaxis?	**Rifampicin 20 mg/kg once daily for 4 days** (max 600 mg) when there is an unvaccinated or incompletely vaccinated child under 4 in the household.
18	Commonest long-term complication of bacterial meningitis and the follow-up?	**Sensorineural hearing loss** (most after pneumococcal meningitis). NICE NG240: **audiology within 4 weeks** of being well enough, preferably before discharge.
19	Subdural effusion after meningitis: clues?	Infant with **persistent or recurrent fever**, **enlarging head circumference**, bulging fontanelle, seizures or focal signs, typically after Hib or pneumococcal meningitis. Cranial US (open fontanelle) or CT.
20	TB meningitis regimen in Uganda and WHO 2022?	UCG 2023: **2RHZE/10RH** (12 months) + steroids. WHO 2022 children's guideline: alternative **6-month intensive regimen (6HRZEto)** for TBM without suspected drug resistance.
21	Steroid dose in TB meningitis?	**Dexamethasone 0.6 mg/kg/day** (WHO Pocket Book) or **prednisolone 2 mg/kg/day (max 60 mg) for 4 weeks**, then taper over 2–4 weeks.
22	MRC staging of TB meningitis?	**Stage I**: alert, no focal signs. **Stage II**: GCS 11–14 or focal signs (e.g. cranial nerve palsy). **Stage III**: GCS ≤10. Higher stage = worse outcome.
23	Cryptococcal meningitis induction (WHO 2022)?	**Single high-dose liposomal amphotericin B 10 mg/kg** + **flucytosine 100 mg/kg/day** + **fluconazole 12 mg/kg/day** (max 800 mg) for 14 days; therapeutic LPs for high pressure; **delay ART 4–6 weeks**.
24	How do you manage raised CSF pressure in cryptococcal meningitis?	**Therapeutic lumbar punctures**: drain CSF until the opening pressure halves or is <20 cm H2O, repeated daily while symptomatic. No steroids, acetazolamide or mannitol.
25	Why measure blood glucose immediately before LP?	To calculate the **CSF:blood glucose ratio**; a CSF glucose that looks normal may be low relative to a high blood glucose (e.g. after dextrose).
26	Fluids in bacterial meningitis?	Give **full maintenance** isotonic fluids (100/50/20 mL/kg/day), NGT feeds when safe. Restrict (to about two-thirds) only with **SIADH** (Na <130, not dehydrated) or raised ICP; never restrict in shock.
27	Effect of antibiotics given before the LP?	CSF **cells, protein and glucose** are little affected in the first ~24 h, but **culture and Gram stain** are often negative. Antigen/PCR tests help.
28	Best LP site in a child?	**L3/4 or L4/5** interspace, on the line joining the iliac crests; in infants use **L4/5** (the cord ends lower). Child lying on the side, held curled, or sitting for neonates.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 7-month-old boy (7.2 kg) presents to Mulago with 2 days of fever, refusal to breastfeed and irritability. He had one generalised seizure lasting 3 minutes at home. He has received pentavalent 1 and 2 only. Temperature 39.0 °C, HR 170, RR 44, CRT 2 s, SpO2 96%. He is irritable, the fontanelle is bulging, there is no neck stiffness, no rash, and pupils are equal and reactive. Malaria RDT is positive. Glucose 4.8 mmol/L. What is the MOST appropriate next step?

A Treat as malaria with IV artesunate and observe because the RDT explains the seizure
B Lumbar puncture, then ceftriaxone 100 mg/kg IV (plus artesunate) without delay
C CT head before any lumbar puncture
D Diagnose a simple febrile convulsion and give paracetamol
E Give ceftriaxone only if the CSF Gram stain is positive

2. A 4-year-old girl (15 kg) arrives with 12 hours of fever and vomiting. She is drowsy (GCS 8), with HR 52 and BP 128/84, irregular breathing, and the left pupil is larger and sluggish. There is no rash. What is the BEST immediate management?

A Immediate lumbar puncture to confirm meningitis before antibiotics
B Lumbar puncture after a 20 mL/kg saline bolus
C Oral amoxicillin and observation for 24 hours
D Secure the airway, head up 30 degrees, ceftriaxone 1.5 g IV now, consider mannitol, and defer the lumbar puncture
E IV dexamethasone alone and await CT before any antibiotic

3. A 3-year-old boy (13 kg) has CSF results: turbid; WBC 1,800/µL (85% neutrophils); protein 1.9 g/L; glucose 1.0 mmol/L with blood glucose 5.5 mmol/L; Gram stain shows Gram-negative diplococci. Which is the likeliest organism?

A Neisseria meningitidis
B Streptococcus pneumoniae
C Haemophilus influenzae type b
D Mycobacterium tuberculosis
E Cryptococcus neoformans

4. A 12-day-old girl (2.9 kg, term, birth weight 3.1 kg) is admitted with poor feeding, temperature 38.4 °C, lethargy and two episodes of apnoea. She is jaundiced (serum bilirubin 210 µmol/L). CSF: WBC 650/µL (70% neutrophils), protein 2.4 g/L, glucose ratio 0.2, Gram-negative rods. IV fluids running are Ringer's lactate with dextrose. Which antibiotic plan is MOST appropriate?

A Ceftriaxone 290 mg IV once daily via the same line
B Chloramphenicol 25 mg/kg 6-hourly
C Oral amoxicillin for 7 days
D Benzylpenicillin alone for 14 days
E Cefotaxime 50 mg/kg IV 8-hourly plus ampicillin (and gentamicin per unit protocol) for at least 21 days

5. A 2-year-old (12 kg) had a 4-minute generalised tonic–clonic seizure with a temperature of 39.5 °C. He has had a runny nose for 2 days. Thirty minutes after the seizure he is playing, alert, with a soft fontanelle and no neck stiffness. He has had no antibiotics. His immunisations are complete including PCV and pentavalent. Malaria RDT negative. What is the MOST appropriate plan?

- A Lumbar puncture and ceftriaxone
- B Start phenobarbital maintenance
- C Treat the fever, find the source (URTI), counsel the parents and observe; lumbar puncture is not needed
- D CT scan of the brain
- E Admit for 5 days of IV antibiotics

6. A 5-year-old boy (18 kg) with suspected pneumococcal meningitis arrives at a Kampala private hospital 6 hours after onset of illness. He is HIV-negative, well nourished, and has had no antibiotics. Your consultant follows NICE NG240. Regarding dexamethasone, what is the correct approach?

- A Give dexamethasone 0.6 mg/kg/day for 4 weeks
- B Give dexamethasone 2.7 mg IV with or just before the first ceftriaxone dose, then 6-hourly for 4 days
- C Give dexamethasone only after 48 hours if hearing loss develops
- D Never give dexamethasone to children under 12 years
- E Give dexamethasone and delay ceftriaxone until the steroid has run in for 2 hours

7. A 3-year-old girl (12 kg) has had fever and vomiting for 14 days. Her grandmother, who lives with her, has a chronic cough. She is drowsy (GCS 12) with a left VI nerve palsy. She has no BCG scar. CSF: clear, WBC 220/µL (85% lymphocytes), protein 2.6 g/L, glucose 0.9 mmol/L (blood 5.0). Gram stain negative. What is the MOST appropriate management?

- A Ceftriaxone for 14 days and repeat LP
- B Aciclovir for presumed herpes encephalitis
- C Fluconazole for cryptococcal meningitis
- D Start anti-TB treatment with steroids now (while sending CSF GeneXpert and a chest X-ray)
- E Wait for CSF TB culture before starting treatment

8. A 13-year-old boy (30 kg) with perinatally acquired HIV stopped ART a year ago. He has had headache for 2 weeks and vomiting for 3 days. GCS 15, no focal signs, mild neck stiffness. Serum CrAg positive. CSF opening pressure 34 cm H2O; WBC 8/µL; protein 0.6 g/L; India ink shows encapsulated yeasts. Which plan is CORRECT?

- A Liposomal amphotericin B 10 mg/kg once plus flucytosine and fluconazole for 14 days, therapeutic LPs, and restart ART after 4–6 weeks
- B Restart ART today and give fluconazole 200 mg daily
- C Dexamethasone and mannitol to lower intracranial pressure
- D Ceftriaxone for 14 days because the neck is stiff
- E Start TB treatment because the CSF protein is raised

9. A 6-year-old (20 kg) is admitted with meningococcal meningitis from a school in Arua. She has a 2-year-old brother and her parents at home. Which is the MOST appropriate prophylaxis for her brother according to UCG 2023?

- A No prophylaxis is needed for household contacts
- B Amoxicillin 25 mg/kg 8-hourly for 5 days
- C Benzylpenicillin IM single dose
- D Rifampicin 20 mg/kg once daily for 4 days
- E Ciprofloxacin 10 mg/kg as a single oral dose

10. A 9-month-old girl (7 kg) on day 8 of ceftriaxone for pneumococcal meningitis develops a new fever of 38.8 °C and two focal seizures. Her head circumference has increased from 43 cm on admission to 45 cm, and the fontanelle is tense. She is otherwise feeding. What is the MOST appropriate next investigation?

A Repeat malaria RDT
B Urine culture
C Cranial ultrasound through the anterior fontanelle
D EEG
E Stop ceftriaxone and observe for drug fever

11. Which CSF finding is NORMAL for a 5-day-old term baby but would be abnormal in a 2-year-old?

A Neutrophils 300/µL
B WBC 15/µL with protein 0.9 g/L
C CSF:blood glucose ratio 0.25
D Positive Gram stain
E Opening pressure 40 cm H2O

12. A 2-year-old (11 kg) with fever and drowsiness has a spreading non-blanching purpuric rash over the legs and trunk, cold hands, CRT 4 s, HR 180, BP 70/40. Platelets are 45 x 10^9/L. Which sequence is BEST?

A Lumbar puncture, then antibiotics, then fluids
B Wait for blood culture results before antibiotics
C Oral ciprofloxacin and discharge with review in 24 hours
D IV/IO access, ceftriaxone 1.1 g immediately, 10–20 mL/kg crystalloid bolus with reassessment, glucose check; no lumbar puncture now
E Platelet transfusion first, then lumbar puncture, then antibiotics

13. A 4-year-old boy (16 kg) with bacterial meningitis on day 2 has serum sodium 126 mmol/L, is clinically euvolaemic (moist mucosa, CRT 2 s, normal BP, passing concentrated urine) and has had no seizures. He is on full maintenance IV fluids. What is the BEST fluid adjustment?

A Reduce isotonic fluids to about two-thirds of maintenance and recheck sodium in 6–12 hours
B Increase fluids to 1.5 times maintenance
C Switch to 5% dextrose in water
D Give 20 mL/kg 0.9% saline bolus
E Give 3% saline 5 mL/kg immediately

14. A 10-month-old (8 kg) has been treated for 10 days for Hib meningitis and is now well. The mother asks what follow-up is needed. Which is the MOST important test to arrange before or soon after discharge?

A Repeat lumbar puncture to confirm sterile CSF
B Serum ferritin
C EEG in all survivors
D Repeat blood culture at 2 weeks
E Hearing assessment (OAE/ABR) within 4 weeks

15. A HC III in Kole district receives a 3-year-old girl (14 kg) with fever, neck stiffness and drowsiness (responds to voice). There is no laboratory for CSF and the hospital is 40 km away. What should the clinician do before referral?

A Perform an LP and send the CSF with the patient
B Give oral amoxicillin and refer
C Give ceftriaxone 1.4 g IM (or IV), check glucose, manage fever and refer urgently
D Give IV fluids 20 mL/kg and refer
E Observe for 6 hours and refer if worse

Answers & explanations

1. B · Lumbar puncture, then ceftriaxone 100 mg/kg IV (plus artesunate) without delay

B is correct: irritability, poor feeding and a **bulging fontanelle** in an infant are signs of meningitis; a positive RDT does not exclude it. He has no contraindication (no shock, no coma, no focal signs, no purpura), so do the **LP now and start ceftriaxone 100 mg/kg (720 mg) within the hour**, while also treating malaria. A is wrong: parasitaemia is common and meningitis often coexists. C is wrong: NICE advises against routine imaging before LP and CT is rarely available; he has no signs of raised ICP warranting delay. D is wrong: a bulging fontanelle and irritability are not features of a simple febrile convulsion. E is wrong: the Gram stain is negative in up to 60% of bacterial meningitis; treat on clinical and CSF cell grounds.

2. D · Secure the airway, head up 30 degrees, ceftriaxone 1.5 g IV now, consider mannitol, and defer the lumbar puncture

D is correct: GCS ≤9, **bradycardia with hypertension and irregular breathing (Cushing triad)** and an **unequal sluggish pupil** mean raised ICP with impending herniation, an absolute contraindication to LP. Protect the airway, nurse head-up, give **ceftriaxone 100 mg/kg (1.5 g)** immediately, consider an osmotic agent (mannitol 0.25–0.5 g/kg) and seek critical care; tap later when safe. A is wrong: LP now risks coning. B is wrong: she is not in shock and a bolus may worsen cerebral oedema; it does not make LP safe. C is wrong: oral antibiotics are inadequate for suspected bacterial meningitis. E is wrong: antibiotics must not be delayed for imaging or steroids.

3. A · Neisseria meningitidis

A is correct: a neutrophilic CSF with high protein and low glucose ratio (1.0/5.5 = 0.18) is bacterial, and **Gram-negative diplococci** are typical of ***N. meningitidis***. B is wrong: pneumococci are **Gram-positive** diplococci. C is wrong: Hib appears as Gram-negative **coccobacilli/rods**, not diplococci. D is wrong: TB gives a lymphocytic CSF and is acid-fast, not Gram-negative. E is wrong: Cryptococcus is a yeast seen on India ink, with few cells.

4. E · Cefotaxime 50 mg/kg IV 8-hourly plus ampicillin (and gentamicin per unit protocol) for at least 21 days

E is correct: Gram-negative neonatal meningitis needs a third-generation cephalosporin that crosses the blood–brain barrier, given for **at least 21 days**; **cefotaxime** (8-hourly after the first week) is preferred, with ampicillin (and often gentamicin) until sensitivities return. A is wrong: ceftriaxone is avoided in a **jaundiced neonate** (bilirubin displacement) and must never run with **calcium-containing Ringer's lactate**. B is wrong: chloramphenicol causes **grey baby syndrome** under 1 month. C is wrong: oral treatment is inadequate for meningitis. D is wrong: penicillin does not cover Gram-negative rods.

5. C · Treat the fever, find the source (URTI), counsel the parents and observe; lumbar puncture is not needed

C is correct: this is a **simple febrile convulsion**: age 6 months–5 years, single brief generalised seizure, **fully alert within an hour**, a clear viral source, fully vaccinated, no prior antibiotics and no meningism. Manage fever, treat the cause, counsel and observe. A is wrong: LP is indicated for complex seizures, slow recovery, meningism, age under ~12 months with doubt, or prior antibiotics, none of which apply. B is wrong: anticonvulsant prophylaxis is not indicated for febrile convulsions. D is wrong: imaging is not needed for a simple febrile convulsion. E is wrong: there is no evidence of bacterial infection.

6. B · Give dexamethasone 2.7 mg IV with or just before the first ceftriaxone dose, then 6-hourly for 4 days

B is correct: NICE recommends IV dexamethasone for children **over 3 months** with strongly suspected bacterial meningitis: **0.15 mg/kg (18 × 0.15 = 2.7 mg) 6-hourly for 4 days**, the first dose **with or before** the first antibiotic, continued only if the organism is pneumococcus or Hib. A is wrong: 0.6 mg/kg/day for weeks is the **TB meningitis** regimen. C is wrong: steroids work only by blunting the early inflammatory burst; late use after deafness has no benefit. D is wrong: age over 3 months is the NICE criterion. E is wrong: antibiotics must never be delayed for steroids.

7. D · Start anti-TB treatment with steroids now (while sending CSF GeneXpert and a chest X-ray)

D is correct: a **subacute** course, **TB contact**, no BCG, **VI nerve palsy** (stage II) and a lymphocytic CSF with **very high protein and very low glucose** (ratio 0.18) = **TB meningitis**. Start TB treatment (UCG 2RHZE/10RH, or the WHO 2022 6-month intensive regimen where adopted) and **steroids** today. A is wrong: bacterial meningitis gives neutrophils and an acute course. B is wrong: HSV gives normal CSF glucose and an acute picture. C is wrong: no HIV history, and crypto CSF usually has few cells. E is wrong: TB culture takes weeks; delay worsens death and disability.

8. A · Liposomal amphotericin B 10 mg/kg once plus flucytosine and fluconazole for 14 days, therapeutic LPs, and restart ART after 4–6 weeks

A is correct: **cryptococcal meningitis** (CrAg, India ink, high opening pressure) needs WHO 2022 induction (**single-dose liposomal amphotericin B 10 mg/kg + flucytosine 100 mg/kg/day + fluconazole 12 mg/kg/day** for 14 days), **therapeutic LPs** for pressure >25 cm H2O, and **ART delayed 4–6 weeks** (early ART increases mortality). B is wrong: immediate ART risks fatal IRIS and fluconazole 200 mg is a maintenance dose. C is wrong: steroids worsen outcomes in crypto and mannitol is not used; CSF drainage is the treatment. D is wrong: few cells and yeasts show this is not bacterial. E is wrong: the protein is only mildly raised and yeasts are visible.

9. E · Ciprofloxacin 10 mg/kg as a single oral dose

E is correct: UCG 2023 recommends **ciprofloxacin** for close contacts of meningococcal disease (especially under 5s): **10 mg/kg single dose** for children under 12 (max 500 mg); alternatives are ceftriaxone single IM dose or rifampicin 10 mg/kg 12-hourly for 2 days. A is wrong: household contacts have a much higher risk and need prophylaxis. B is wrong: amoxicillin does not eradicate nasopharyngeal carriage. C is wrong: penicillin treats disease but does not reliably clear carriage. D is wrong: 20 mg/kg daily for 4 days is the **Hib** prophylaxis regimen.

10. C · Cranial ultrasound through the anterior fontanelle

C is correct: new fever, focal seizures, a **2 cm increase in head circumference** and a tense fontanelle after pneumococcal meningitis suggest **subdural effusion/empyema or hydrocephalus**; with an open fontanelle, **cranial ultrasound** is quick, safe and available at most Ugandan hospitals (CT if not). A is wrong: malaria does not explain the enlarging head. B is wrong: UTI does not explain focal seizures and head growth. D is wrong: EEG does not identify the structural cause. E is wrong: drug fever is a diagnosis of exclusion; this picture needs imaging and possibly neurosurgical drainage.

11. B · WBC 15/µL with protein 0.9 g/L

B is correct: neonatal CSF normally has **up to about 20 WBC/µL** and **protein up to 1.0 g/L**, whereas after 1 month the limits are **≤5 WBC/µL and protein <0.4 g/L**. A is wrong: 300 neutrophils is abnormal at any age. C is wrong: a ratio below 0.4 is abnormal at any age (normal ≥0.6). D is wrong: a positive Gram stain is never normal. E is wrong: 40 cm H2O is markedly raised at any age.

12. D · IV/IO access, ceftriaxone 1.1 g immediately, 10–20 mL/kg crystalloid bolus with reassessment, glucose check; no lumbar puncture now

D is correct: this is **meningococcal septicaemia with shock**: give **ceftriaxone 100 mg/kg (1.1 g)** at once, resuscitate with 10–20 mL/kg boluses and reassess, check glucose, and seek critical care. **Shock, extensive purpura and low platelets** all contraindicate LP. A is wrong: LP is contraindicated and would delay life-saving treatment. B is wrong: antibiotics must not wait for cultures. C is wrong: this child is critically ill. E is wrong: correcting platelets to perform an LP delays antibiotics; LP is not needed to start treatment.

13. A · Reduce isotonic fluids to about two-thirds of maintenance and recheck sodium in 6–12 hours

A is correct: hyponatraemia with euvolaemia and concentrated urine in meningitis suggests **SIADH**; the guideline approach is to **restrict isotonic fluids to about two-thirds of maintenance** and monitor sodium. B is wrong: more free water worsens hyponatraemia and cerebral oedema. C is wrong: hypotonic dextrose-water lowers sodium further. D is wrong: he is not shocked; a bolus adds volume without correcting SIADH. E is wrong: hypertonic saline is for **symptomatic** (seizures) or severe (<120 mmol/L) hyponatraemia.

14. E · Hearing assessment (OAE/ABR) within 4 weeks

E is correct: **sensorineural hearing loss** is the commonest sequela of bacterial meningitis (especially Hib and pneumococcus); NICE NG240 advises **audiology within 4 weeks**, and early detection allows hearing aids or cochlear implant referral before ossification. A is wrong: a test-of-cure LP is not needed in an uncomplicated recovering child. B is wrong: iron status is not a priority follow-up. C is wrong: EEG is only for children with ongoing seizures. D is wrong: repeat blood culture is not needed after clinical cure.

15. C · Give ceftriaxone 1.4 g IM (or IV), check glucose, manage fever and refer urgently

C is correct: UCG 2023 says refer all suspected meningitis **after a pre-referral dose of antibiotic**: **ceftriaxone 100 mg/kg (1.4 g)** IM or IV, plus checking glucose, treating fever and seizures, and urgent referral. A is wrong: LP without lab support delays treatment and may be unsafe at HC III. B is wrong: oral amoxicillin does not achieve CSF levels. D is wrong: she is not shocked; boluses are not indicated. E is wrong: delay increases death and disability.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: An irritable 8-month-old with a bulging fontanelle

You are the intern on call at Mengo Hospital. An 8-month-old boy from Wakiso is brought at 14:00 with 2 days of fever, refusal to breastfeed and irritability. Thirty minutes ago he had a seizure of the left arm lasting 10 minutes. Weight 7.6 kg. His card shows BCG, OPV0–3, pentavalent 1–2, PCV 1–2; he missed the 14-week visit.

1. What are your priorities in the first 5 minutes?

> ETAT: airway, breathing (RR 44, SpO2 96%), circulation (HR 168, CRT 2 s, warm: not shocked), consciousness (irritable, responds to voice: AVPU V), not convulsing now. **Glucose 4.6 mmol/L**, RDT, blood culture, FBC, electrolytes. Examine for rash, fontanelle (bulging), pupils, focal signs, neck.

2. Is an LP safe? What will you check first?

> Check for **contraindications**: no shock, no respiratory failure, seizure has stopped, no extensive purpura, platelets (FBC) if available, no focal neurology now, pupils equal and reactive, conscious level not falling, no skin infection over the back. A focal seizure that has stopped with full recovery is not by itself a contraindication. **LP is reasonable now** if it does not delay antibiotics beyond the first hour; measure blood glucose just before.

3. CSF is turbid: WBC 2,400/µL (90% neutrophils), protein 2.1 g/L, glucose 0.8 mmol/L (blood 5.2), Gram-positive diplococci. Interpret and treat.

> **Acute bacterial meningitis, probably pneumococcal** (ratio 0.15). **Ceftriaxone 100 mg/kg = 760 mg IV once daily** (or 380 mg 12-hourly) for 10–14 days. Dexamethasone: per unit policy (NICE would give 0.15 mg/kg = 1.1 mg 6-hourly × 4 days with the first antibiotic; WHO/UCG do not recommend routinely). Paracetamol 15 mg/kg = 115 mg; diazepam 0.5 mg/kg PR if further seizures, phenobarbital 15–20 mg/kg if recurrent.

4. Outline his supportive care and monitoring.

> **Maintenance fluids** 100 mL/kg/day = 760 mL/day (≈32 mL/h), mostly as breast milk/NGT feeds if safe; **daily sodium** (SIADH); **head circumference daily** (baseline today); GCS/AVPU, pupils, vitals 4-hourly; turn and nurse head-up 30°; glucose checks. **Notify** the district surveillance focal person.

5. On day 7 he becomes febrile again and the head circumference has increased by 1.5 cm. What do you suspect and do?

> **Subdural effusion/empyema**, **hydrocephalus** or **abscess** (also consider phlebitis, drug fever, nosocomial infection). **Cranial ultrasound** via the open fontanelle (CT if unclear), repeat LP if the ultrasound is unhelpful, neurosurgical opinion if empyema or obstructive hydrocephalus. Continue antibiotics; consider extending duration.

6. What will you arrange at discharge?

> **Hearing assessment within 4 weeks** (OAE/ABR at Mulago/Mengo ENT), **developmental follow-up** (milestones, tone, vision), seizure plan, **catch-up pentavalent 3 and PCV 3** today, measles-rubella at 9 months, HIV test if not documented, growth monitoring, and danger signs for return.

Take-away: In an infant, irritability, poor feeding and a bulging fontanelle are meningitis signs; tap when safe, treat within the hour, measure the head daily and never forget the hearing test.

Case 2: A lethargic 10-day-old on the neonatal unit

A 10-day-old girl, born at term at a HC IV (birth weight 3.1 kg), is brought to the Mengo neonatal unit with 1 day of poor feeding and 'sleepiness'. Her mother had prolonged rupture of membranes (30 hours) and fever in labour. Today the baby weighs 2.9 kg.

1. What specific findings will you look for?

> Temperature (38.4 °C or hypothermia), lethargy, poor suck, **high-pitched cry**, **apnoea** or desaturation, **bulging fontanelle**, seizures (subtle: lip-smacking, cycling, eye deviation), jaundice, capillary refill, glucose, signs of other foci (umbilicus, skin, chest). Weigh (2.9 kg = 6% below birth weight).

2. She has two apnoeic episodes and is jaundiced (bilirubin 210 µmol/L). Glucose 2.1 mmol/L. What immediate care?

> Stimulate/bag and mask as needed, oxygen to SpO2 ≥92%, warm (incubator/radiant warmer), **10% dextrose 2 mL/kg = 6 mL IV**, then maintenance IV 10% dextrose (about 150 mL/kg/day at day 10) or NGT expressed breast milk if tolerated. Blood culture, FBC, CRP. **LP** once breathing and glucose are stable.

3. CSF: WBC 650/µL (70% neutrophils), protein 2.4 g/L, glucose ratio 0.2, Gram-negative rods. Prescribe.

> **Cefotaxime 50 mg/kg = 145 mg IV 8-hourly** (age >7 days) **plus ampicillin 100 mg/kg = 290 mg IV 8-hourly** (± gentamicin per protocol: WHO 7.5 mg/kg daily after week 1; UCG 2.5 mg/kg 12-hourly), adjusting to culture. **Avoid ceftriaxone** (jaundice, calcium-containing fluids). Duration **at least 21 days**. Seizures: **phenobarbital 20 mg/kg = 58 mg IV** loading.

4. Culture grows Klebsiella resistant to cefotaxime (ESBL). What now?

> Switch to **meropenem 40 mg/kg IV 8-hourly** (meningitis dose) for at least 21 days; repeat LP at 48–72 h to confirm sterilisation; monitor for ventriculitis/abscess with cranial ultrasound. Report to infection control (ESBL outbreaks in neonatal units).

5. What follow-up does she need?

> **Hearing screen (ABR)**, cranial ultrasound before discharge, **head circumference** at every visit, neurodevelopmental follow-up (cerebral palsy, epilepsy, delay), breastfeeding support and weight gain, immunisations (BCG, OPV, HepB birth dose if missed), and maternal counselling about early signs of illness.

Take-away: Neonatal meningitis looks like 'not feeding well and sleepy'; LP every septic neonate when stable, use ampicillin plus cefotaxime (not ceftriaxone with jaundice or calcium) and treat Gram-negative meningitis for at least 21 days.

Case 3: Headache in an adolescent with HIV: crypto or TB?

A 13-year-old boy (30 kg, BMI-for-age Z −2.2) with perinatally acquired HIV comes to Mulago paediatric emergency with 2 weeks of headache, 3 days of vomiting and blurred vision. He stopped ART a year ago. His mother was treated for TB 6 months ago.

1. What is your differential diagnosis?

> **Cryptococcal meningitis**, **TB meningitis** (household contact), bacterial meningitis (less likely with 2 weeks), toxoplasmosis or other space-occupying lesion (CNS lymphoma rare), HIV encephalopathy, cerebral malaria (fever?).

2. Examination: GCS 15, mild neck stiffness, no focal signs, bilateral papilloedema. What tests, and is LP safe?

> **Serum CrAg** (positive), RDT, FBC, electrolytes, creatinine, CD4, viral load later, chest X-ray, sputum/gastric Xpert. Papilloedema in crypto is usually from communicating high CSF pressure; with **no focal signs and normal consciousness**, many guidelines proceed with LP **with opening pressure**, but if focal signs or reduced GCS, image first (CT) to exclude a mass lesion. Discuss with the senior.

3. LP: opening pressure 36 cm H2O, WBC 8/µL, protein 0.6 g/L, glucose 2.6 mmol/L (blood 5.0), India ink positive, CSF CrAg positive, Xpert negative. Treatment?

> **Cryptococcal meningitis.** Induction: **liposomal amphotericin B 10 mg/kg = 300 mg IV once** + **flucytosine 100 mg/kg/day = 750 mg 6-hourly** + **fluconazole 12 mg/kg/day = 360 mg daily** for 14 days (WHO 2022); then fluconazole 6–12 mg/kg/day for 8 weeks, then 6 mg/kg/day (max 200 mg). **Therapeutic LP**: drain CSF until pressure <20 cm H2O or halved; repeat daily while symptomatic. Pre-hydration and potassium with amphotericin; monitor creatinine and K+.

4. When will you restart ART and what else is needed?

> **Delay ART 4–6 weeks** after starting antifungals (early ART increases mortality; UCG flowchart). Start **cotrimoxazole**. Screen for TB thoroughly (given the contact); if TB is diagnosed, treat it too and plan ART timing with the ART clinic. Enhanced adherence counselling and psychosocial support for an adolescent who stopped ART; disclosure and school issues.

5. How would the CSF have differed if this were TB meningitis?

> TBM: more cells (**lymphocytes 50–500/µL**), **protein 1–5 g/L**, **very low glucose (ratio <0.3)**, cobweb clot; India ink/CrAg negative; possible cranial nerve palsy and hydrocephalus. Treatment would be **2RHZE/10RH (or WHO 2022 6-month intensive regimen) plus steroids**, with ART after 4–8 weeks.

> **Take-away:** In an HIV-positive child with subacute headache, check serum CrAg and the CSF opening pressure; cryptococcal meningitis needs combination antifungal induction, repeated pressure-relieving LPs and delayed ART, while TBM needs TB drugs plus steroids.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.